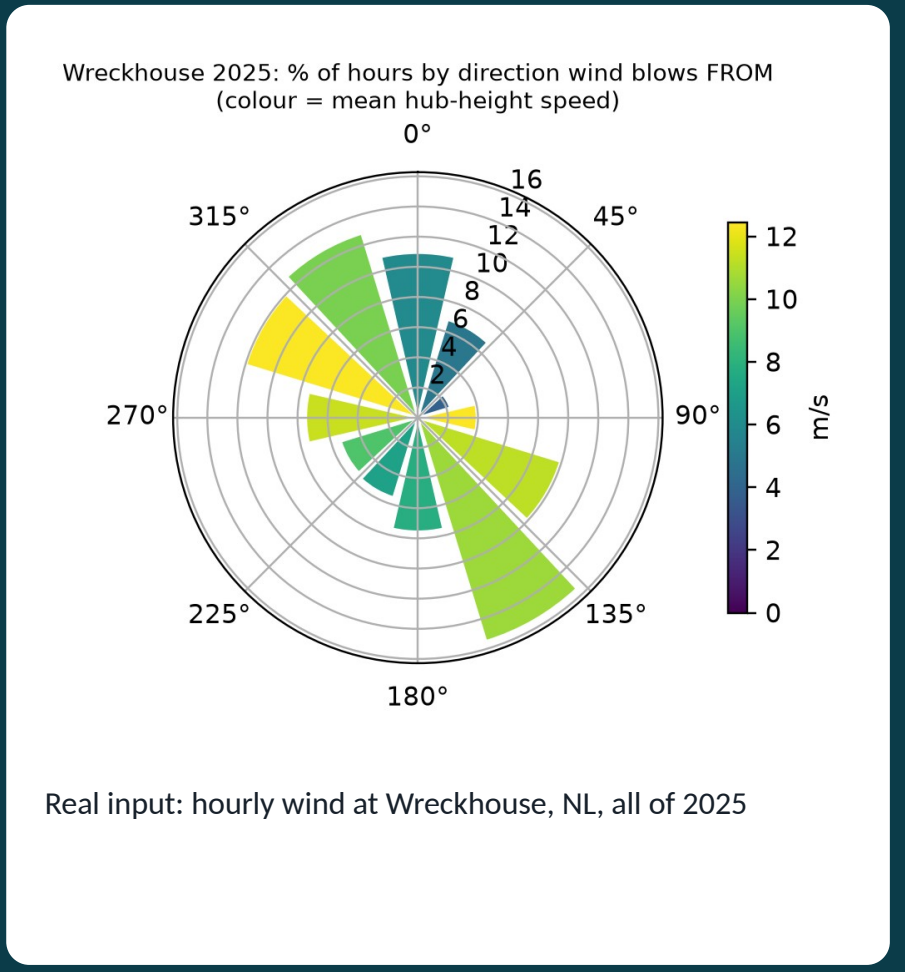
WakeWise NL

Wake-aware wind-turbine siting for Newfoundland's west coast, solved with QAOA in Qiskit

Team AI Entanglers

Qiskit Fall Fest 2026 · Memorial University



Real input: hourly wind at Wreckhouse, NL, all of 2025

Strong NL wind meets a proposed hydrogen farm

Wreckhouse, southwest Newfoundland

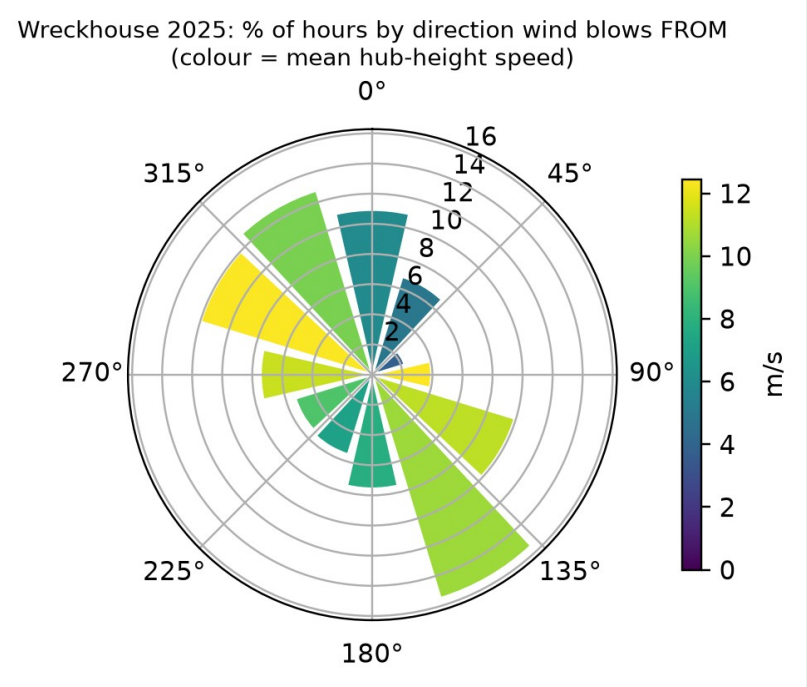
Gales here once blew trains off the tracks. In 2025 the wind at a 110 m hub averaged 9.4 m/s: one 4 MW turbine would run at a 40% capacity factor.

Project Nujio'qonik, next door

150+ turbines proposed on the Port au Port Peninsula and in the Codroy Valley to make hydrogen in Stephenville. Released from provincial environmental assessment, with conditions, in April 2024.

MUN's CREATE-SWIFT program asks for this

Benchmark wind-energy optimization: classical versus quantum or hybrid methods, on accuracy, cost, scalability and real-world relevance.



Bimodal wind: south-easterly gales and north-westerlies. Turbines lined up on that axis shade each other most.

Sources: ECCC hourly climate data (Wreckhouse, ID 8404343); NL Government EA bulletin, 9 Apr 2024; CBC News; CREATE-SWIFT talk, Qiskit Fall Fest 2026.

Wakes waste wind, so siting is a real decision

13,993 MWh/yr

from one 4 MW turbine standing alone at Wreckhouse (real 2025 wind)

up to 4.4%

of that output lost to a single badly placed neighbour's wake

9.6%

of achievable net energy a greedy planner leaves behind on 12 sites

Decision and intended use

Which candidate sites get a turbine (bit = 1) or not (bit = 0). The output is a ranked set of layouts and the energy at stake, for early-stage screening: not construction advice.

Sustainability benefit

More clean electricity, or green hydrogen, from the same land, steel and concrete; or the same energy from fewer turbines and less disturbed ground. Our prototype shows the method; the real benefit needs site-specific terrain, turbines and costs.

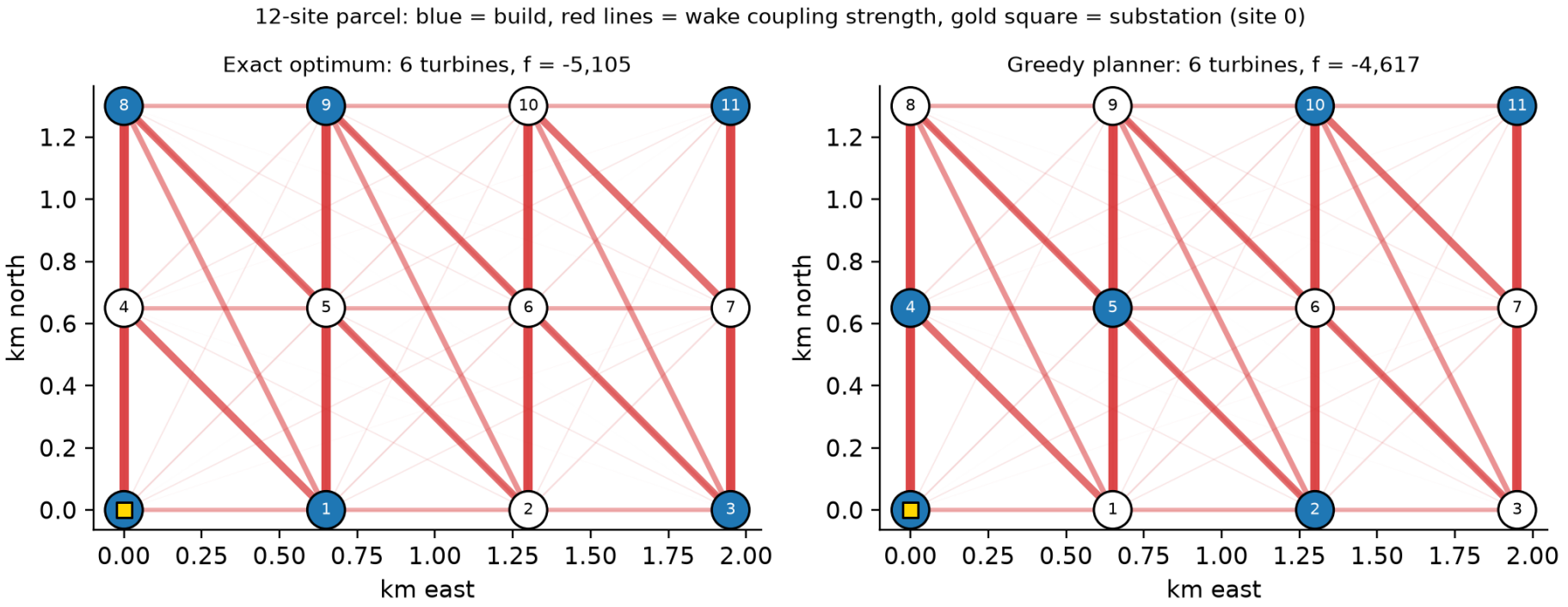
Inputs, output and scope: real wind, illustrative rest

Input	Value used	Status
Hourly wind direction and speed	Wreckhouse, ECCC, 2025 (8,740 hours)	**Observed**
Hub-height wind	10 m to 110 m with power-law shear, alpha = 0.14	Assumption
Turbine	4 MW, 130 m rotor, generic power curve, thrust coefficient 0.8	Illustrative
Wake model	Jensen/Park, decay 0.075, losses added pair by pair	**Standard model, checked**
Parcel	Flat grid, 650 m (5 rotor diameters) apart	Illustrative
Costs, expressed as energy	88% of a turbine's output + 2% per 650 m of cable	Illustrative, sensitivity tested
Output	Build / no-build bitstring = one layout	
Scope	12 sites (12 qubits) in simulation; 9 sites (9 qubits) on noise model and hardware	

Greedy siting leaves 9.6% of the energy behind

$$f(x) = -\Sigma\,(E - c_i)\,x_i \;+\; \Sigma\,W_{ij}\,x_i\,x_j$$

E: one turbine's energy · c_i: annualized cost as energy · W_{ij}: energy lost when both i and j are built (MWh/yr, lower is better)



Baselines on the same instance

- Exact search of all 4,096 layouts
- Greedy: add the best site until none helps
- Uniform random guessing (control)

Greedy gap: 9.6% on 12 sites, 5.9% on 9; 3 to 10% across cost assumptions

The wake physics is already an Ising model

$$x_i = (1 - z_i) / 2$$

$$H = \Sigma\, h_i Z_i + \Sigma\, J_{ij} Z_i Z_j$$

Turbine value becomes a Z field; the wake loss between two turbines becomes a ZZ coupling.

Why QAOA fits

- Natively quadratic and binary: no encoding overhead, no penalty terms
- NP-hard in general (contains maximum-weight independent set)
- Small enough to know the exact answer, so QAOA's quality is measured, not assumed
- Checked: the Ising energy equals f(x) for all 4,096 layouts

1 **Map** *classical*
ECCC wind → wake model → QUBO → SparsePauliOp → qaoa_ansatz, depth p = 1 to 3

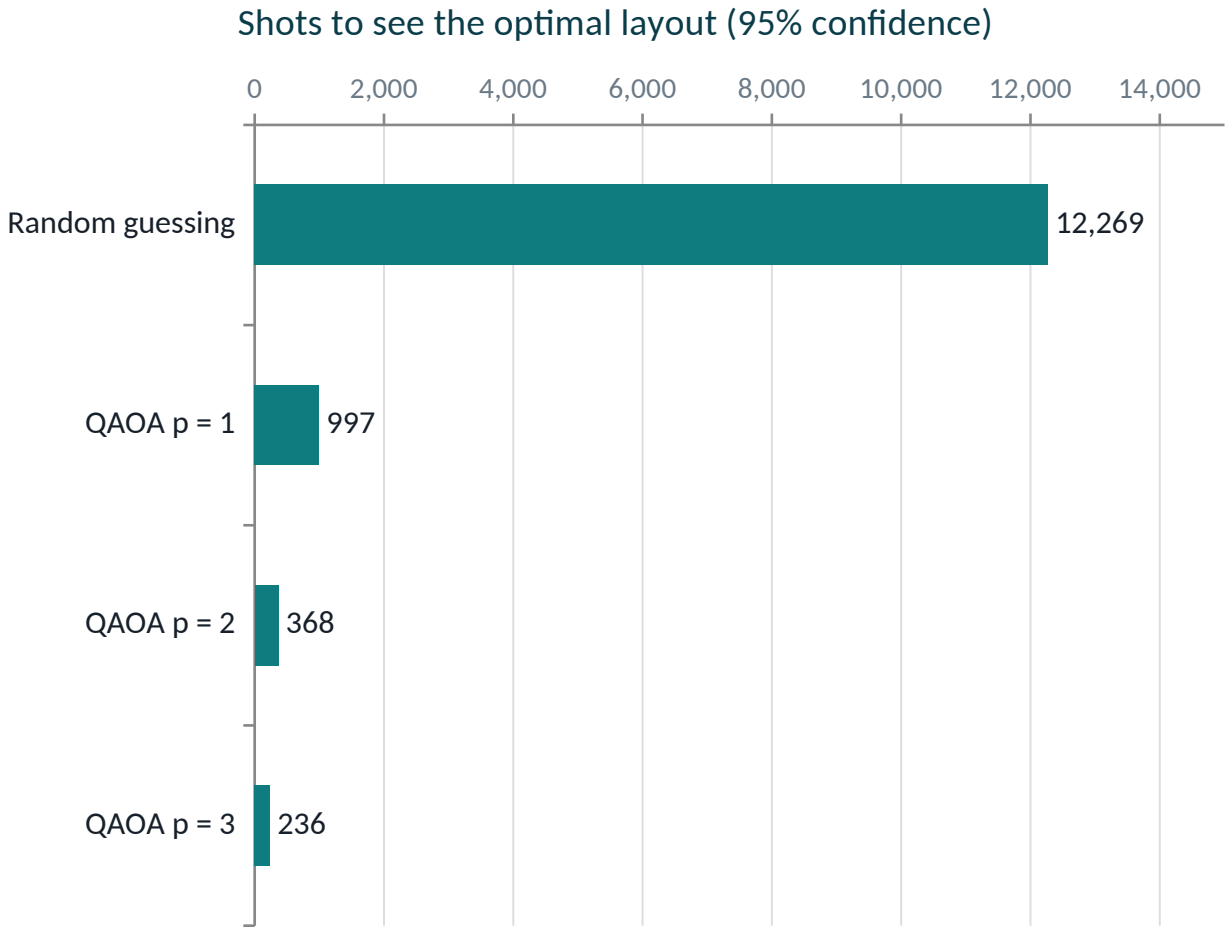
2 **Optimize** *classical*
Transpile for IBM Heron (level 3); drop weak couplings to save two-qubit gates

3 **Execute** *quantum*
Train angles on StatevectorEstimator + COBYLA; sample layouts: ideal, FakeFez noise model, IBM hardware

4 **Post-process** *classical*
Score every sampled layout with the full QUBO; compare with exact, greedy and random

QAOA finds the best layout ~52x more than chance

Noiseless simulation



1.26% vs 0.02%

of shots return the optimal layout (QAOA p = 3 vs random guessing)

72% vs 2%

chance that 100 shots contain the optimal layout

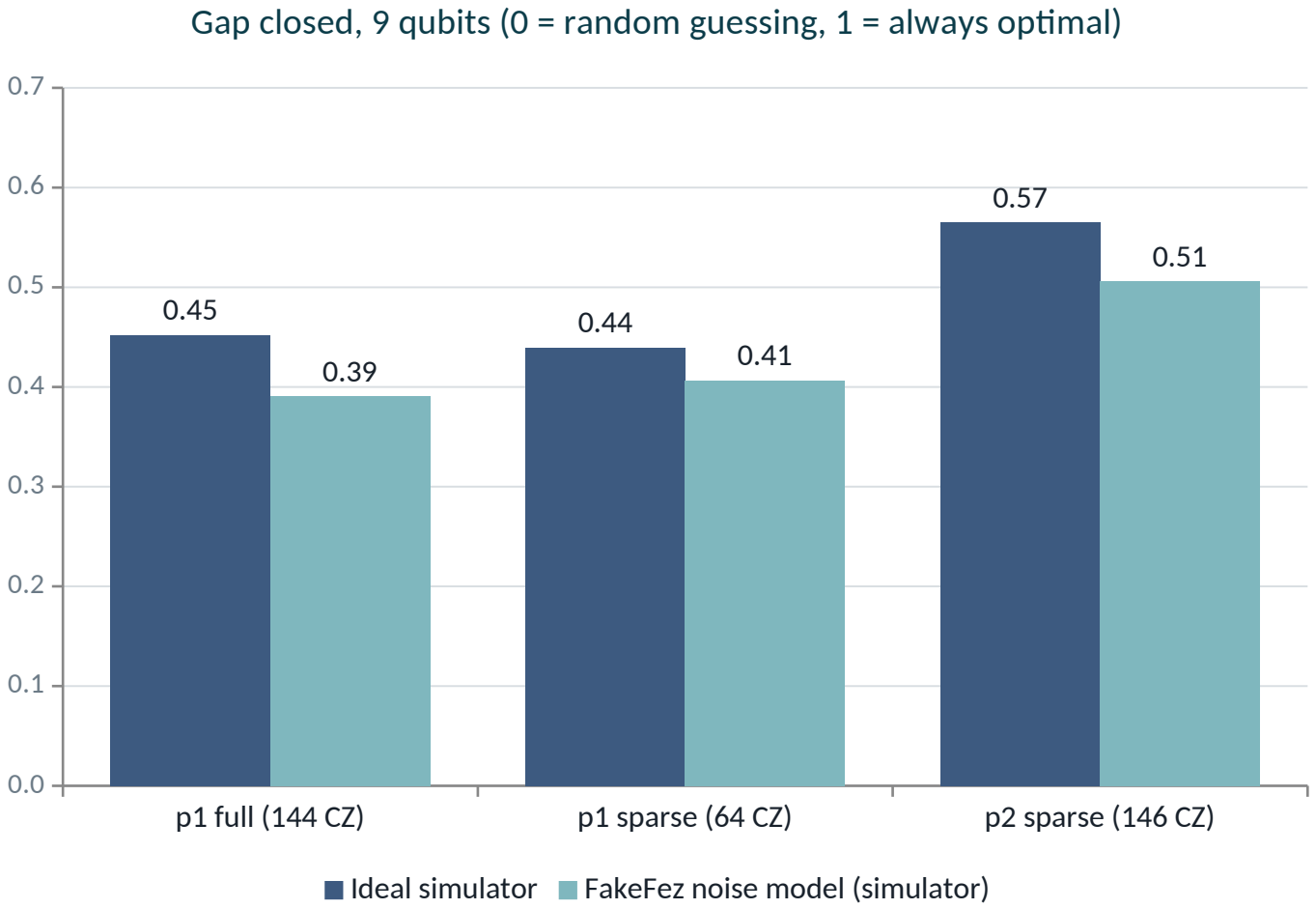
26%

of QAOA p = 3 samples beat the greedy planner's layout

Honest caveat: the average QAOA sample (gap closed 0.62) is still worse than greedy's single answer (0.76). The value is in the best of a few hundred shots.

Under realistic noise, the signal survives

Simulator



9 sites, 4,000 shots per circuit; angles trained in simulation (no QPU time spent on training).

Dropping couplings below 25% of the strongest cuts CZ gates 144 → 64 at p = 1, with little loss in quality.

Hardware run: pending.

Random guessing hits the optimum in 0.20% of shots on 9 sites.

What the evidence supports, and what it does not

Supports

- Siting is genuinely combinatorial on real NL wind: greedy misses 6% to 10% of achievable net energy
- QAOA biases sampling toward good layouts (52x random at p = 3) and improves with depth
- The signal survives a realistic IBM Heron noise model
- The pairwise QUBO ranks layouts like a richer multi-wake model (Spearman 0.99)

Does not support

- Quantum advantage: a laptop checks all 4,096 layouts in milliseconds
- Real-world energy figures: flat idealized parcel, one wind year, illustrative costs
- QAOA beating greedy on a typical sample: only its best samples do

Limitations and next steps

Limitations and resources

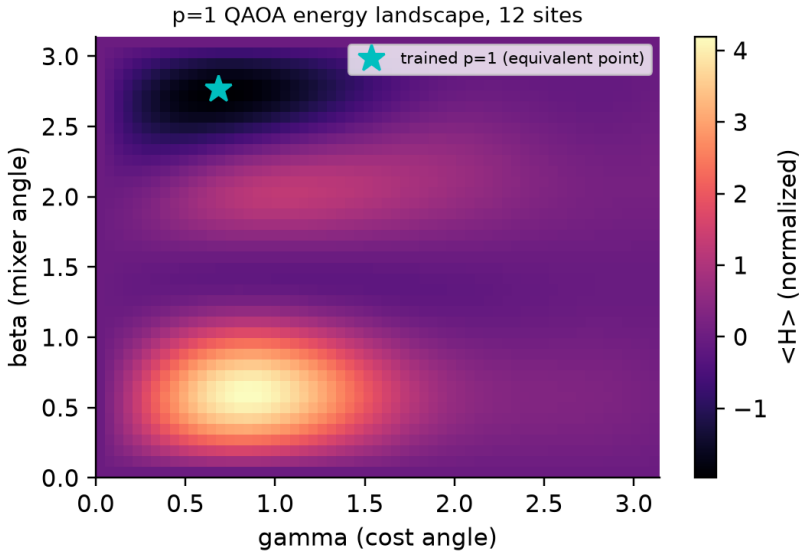
- Model: flat terrain, one wind year, 10 m data extrapolated to hub height, generic turbine, Jensen wake
- Quantum: depth $p \leq 3$, angles trained in simulation; dense couplings need SWAPs on heavy-hex, hence 9 sites on hardware
- Resources: the full notebook runs in about 2 minutes on a laptop

Next steps

1. Real Codroy Valley parcel with terrain-aware wind (50 m reanalysis or WRF)
2. Spacing and exclusion zones (wetlands, caribou habitat) as extra QUBO terms
3. Constraint-preserving mixers, warm starts and CVaR training
4. Scale past exact search; compare with simulated annealing and MILP
5. Partner route: MUN CREATE-SWIFT, with Mitacs Accelerate funding

More clean energy from the same wind

- Real Wreckhouse wind + wake physics gives a QUBO where greedy misses 9.6%
- QAOA in Qiskit finds the optimal layout ~52x more often than chance; the signal survives noise
- Honest scope: a verified proof of concept, no advantage claimed



p = 1 QAOA energy landscape for the 12-site parcel; the star marks the trained angles.

github.com/mrw-soumik/Qiskit-AI-Entanglers-

Thank you. Questions?

Credits: QAOA workflow adapted from Tinh Bui's workshop notebook; Qiskit 101 notebook (IBM); ECCC data (Open Government Licence). Full credits in the notebook.